PROVISIONAL APPLICATION DRAFT - REVISED

Title

AI-CONTROLLED HIGH-PERFORMANCE UNCREWED AIRCRAFT WITH CONFIDENCE-GATED ADAPTIVE FLIGHT MANAGEMENT

Disclosure

A high-performance uncrewed aircraft platform having heterogeneous sensors, onboard compute, confidence-aware state estimation, an environment model with uncertainty, an AI planning engine, deterministic safety gating, health-aware reconfiguration, authenticated human supervision, resilient communications, mission-state reconciliation, and evidence logging.

The platform generates candidate navigation or flight-management actions from mission constraints and current state, evaluates candidate behavior using a forward evaluator, and passes candidate outputs through deterministic protected constraints before flight control.

A predicted health state can update planning constraints before a new trajectory is selected.

Authenticated mission contracts can define authorized action classes, validity, operating constraints, and revocation state.

During degraded communications, the platform can restrict executable behavior to previously authenticated actions and retain a local mission-state record for later reconciliation.

Evidence may include model identity, state snapshots, confidence, health, command provenance, communications state, and safety-envelope decisions.

Optional filing note

This provisional draft is only an early-filing aid. Confirm the disclosure is complete enough for the desired priority claim and use the current USPTO provisional filing procedures.